

Agapet: Supporting A Responsible Adoption Process for Stray and Abandoned Pets in Urban Areas

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ABSTRACT

The growing number of stray domestic animals negatively impacts cities' possibilities to be safe and resilient spaces for their citizens, thus hindering possibilities of attaining the SDG 11 of fostering sustainable cities and communities. Organizations and governments strive to ensure stray pets are adopted. However, doing so is challenging: citizens often do not want to adopt, and many of those who do end up returning their pets to shelters. This situation worsens in countries from the Global South where in-place infrastructures struggle to reinforce existing policies. During four months, we worked closely with a civic-led rescue organization in Ecuador to explore how technology can support organizations in motivating responsible and sustained adoption. As a result, we propose Agapet, a mobile app that, using a timeline, visual representation of milestones, short pet-ownership courses, and a rewards system, sheds light on all the stages of the adoption journey. In this article, we summarize our design process and how the positive results we have obtained so far suggest the app is a step towards responsible pet ownership in urban spaces.

CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

KEYWORDS

stray animals, designing with pet rescue organizations, user-centered design, design journey

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1 INTRODUCTION

Large numbers of stray and abandoned animals in urban areas is an issue that directly hinders cities' possibilities to be safe and resilient spaces for their inhabitants. With no home or caregivers, animals wander around, damaging private property, hindering traffic, and increasing noise pollution [10]. Further, lack of proper care heightens the probability of sanitary emergencies that affect human communities [1]. These negative effects tend to impact countries in the Global South more deeply, as the percentage of stray animals is higher than in other regions of the world, and governmental infrastructure is often weaker or missing [6, 9]. As such, this is a problem critical to address when working towards the **SDG 11: Sustainable Cities and Communities**.

Shelters work hard to rescue these pets and prepare them for adoption. While ensuring timely and responsible adoption is critical, obstacles prevent it from happening as fast and safely as it needs to. Over a third of the population dismisses the idea of adopting due to a preference for particular breeds and concerns about these pets being aggressive, fearful, and destructive due to their background [7]. Even when seemingly successful, the risks of owners returning the pet to shelters due to incompatibilities with the pet's behavior or health issues is quite high (20% of all adoptions) [5]. If adoptions do not occur, shelters quickly run out of resources and cannot continue rescuing pets from city streets. Motivating responsible adoption, thus, becomes vital for ensuring sustainable cities.

For over four months, we followed a user-centered design approach working with a civic-led animal shelter in Guayaquil, Ecuador, to propose a response to this need. After observing their interactions with potential adopters and interviewing volunteers and adopters, we found that a critical factor hindering responsible adoption is the high levels of uncertainty that possible and new adopters feel about the adoption process and the adoption experience. Some of the reasons adopters gave for quitting the process were little understanding of what the adoption process requires, how to be a good owner, and what to expect once the pet arrives.

Based on our findings and the iterative design process, we present Agapet, a mobile app that, using a timeline, visual representation of milestones, short pet-ownership courses, and a rewards system, sheds light on all the stages of the adoption journey. This paper describes existing solutions for responsible pet adoption and explains the need for another form of support that addresses adopters' uncertainties. We then describe our user research and design process, present the current Agapet version, and the results of our latest evaluations about its potential to decrease uncertainty and motivate adoption in countries such as Ecuador. By doing so, the

app supports the city of Guayaquil's work towards the SDG 11 goal of fostering sustainable cities and communities.

2 RELATED WORK

Globally, rescue organizations tend to operate under a limited physical, and financial infrastructure to house stray animals [8]. Even though they often seek the support of private organizations and volunteers, organizations often struggle to secure enough resources. Thus, they need to ensure a constant flux of animals through adoption. To that end, different technologies offer various forms of support to these organizations. Social networks such as Facebook, Instagram, and TikTok enable them to advertise animals for adoption and to organize adoption campaigns in parks, veterinary centers, or pet shops. Organizations also resort to online forms to receive adoption applications from those interested and messaging apps to communicate with potential adopters [12].

With regards to technologies specifically designed to support the adoption process, Pet Finder¹ and Bleppy²—openly accessible in Google Playstore, allow potential adopters to search different types of pets (e.g., dogs, cats) from shelters and find a good match. Other apps such as Anipal, Furescue, and Leen target Global South contexts specifically [2?, 3]. Recognizing organizations' limitations, for example, Anipal attempts to support citizens in doing much of the work organizations do, including reporting stray animals and seeking potential adopters. As such, this app aims at forming a solid network of rescuers and adopters. Leen and Furescue, on the other hand, try to support organizations in finding donations and perfect matches for adoption requests [2, 3].

However, there is a pending need also to increase opportunities for adoption in a responsible and, thus, sustained manner. That is, to last over time. Apps such as 11Pets and Dogo, which are helpful for people who already own a pet, suggest that technology can help to provide owners with knowledge about how to live with their pets. 11Pets allows owners to track the wellness of a pet, using a calendar for baths, vaccines, and clinical history³. Dogo provides owners with resources to learn how to train pets with training routines, a social media component to share experiences and solutions to training problems⁴. These apps highlight the need for more explorations on how technology can help owners to navigate the challenges that an adoption process might entail—before, during, and after.

3 USER RESEARCH

We worked on the problem of motivating responsible pet adoptions with "YoAmoALosAnimales," a civic-led pet-rescuing organization in Guayaquil, Ecuador. They proposed this problem as a project for our HCI undergraduate class. Our approach to exploring the problem area entailed unpacking and contrasting how organizations and adopters defined and experienced the adoption process. To understand the organizations' perspective, we interviewed two administrators and four volunteers from 3 different organizations. We also inspected various organizations' social media accounts, explored their adoption posts and adoption processes, and attended

¹<https://play.google.com/store/apps/details?id=com.discovery.petfinder>

²<https://bleppy.app/>

³<https://www.11pets.com/en>

⁴<https://dого.app/>

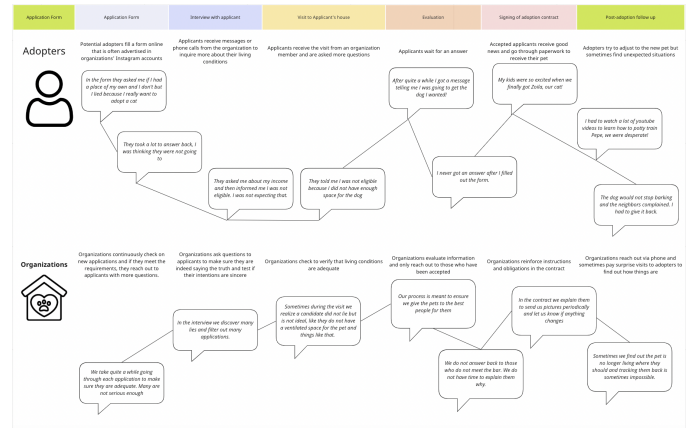


Figure 1: User Journey Map of the adoption process from the perspective of adopters and organization staff

an adoption fair. In total, we conducted 10 hours of observation. To analyze the adopters' view of the process, we interviewed 14 adopters who had both positive and negative adopting experiences. Finally, we visited and observed the households of 5 families who had adopted pets.

4 FINDINGS AND DESIGN IMPLICATIONS

After analyzing the collected data using an Affinity Diagram, we used a user journey map to represent and contrast the different views of the adoption process. As shown in fig1, both organizations and adopters have had particularly negative experiences when waiting for certain events to occur without knowing how they will turn out.

As one of the adopters stated, applying for adoption can feel like "jumping off a cliff" without knowing what organizations expect from applicants. Organizations' staff confirmed that they have a clear list of criteria they are looking for in adopters (e.g., a good, steady income, a place of their own, an appropriate space for the pet, and so on). They see their process as one that filters out inadequate candidates. We found, however, that due to a lack of time and resources, organizations often do not reach out to non-selected applicants. A participant explained, "not knowing what happened is very discouraging. I have almost given up on trying to adopt."

Even those whose application was accepted felt uncertain when the new pet's needs clashed with existing routines and expenses. New owners shared they often felt left alone: "my dog was constantly barking at night, my neighbors were always complaining about it, and there was no one I could ask for advice."

Technology could help keep adopters informed about what to expect through the adoption process. Based on our findings, we propose three design implications where technology use can support the process: (1) to ensure that the adopter always knows their status in the process and what is expected of them throughout; (2) to assist the adopter in learning more about what it means to adopt and own a pet; and (3) to provide immediate guidance in case of doubt without overburdening organization's staff.

5 DESIGN PROCESS

Based on these design implications, we followed an iterative design process in collaboration with our partner organization. We discussed each design and user testing cycle with the organization staff before the next iteration.

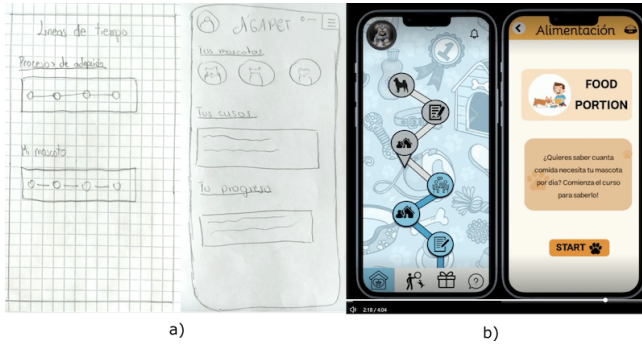


Figure 2: Design iterations: a) Paper Prototype and b) Video Prototype tested at adoption fair

5.1 Ideation and Proposed Concept

Using the derived design implications, we engaged in an ideation process that produced concepts such as an online crash course for adopters to take before applying for adoption and an online social network for experienced and novice adopters. While an online class does attend to adopters' learning needs, mandatory could discourage potential adopters even more. An online social network could offer learning opportunities and immediate support. However, it could deviate from the primary goal of rendering visible the adoption process, including the applicant status within it.

To provide a holistic and less restrictive solution, we proposed Agapet. We thought of it as a mobile app that an organization can recommend to users after applying for adoption. Via Agapet, adopters can clearly see the timeline of their adoption process and and be aware of what is going on and what will happen next. Based on the adopter's status, Agapet suggests short courses to take to prepare for the next stage. Finally, Agapet allows adopters to access immediate assistance via a chatbot. Agapet evolved through several prototyping iterations.

5.2 First Iteration: Paper Prototyping

We paper-prototyped the main functionalities, including the list of courses adopters could take, the adoption timeline, and an online chat box with a bot. See fig2.a. We tested the prototype with five potential adopters recruited from our HCI class. The results indicated that, although participants considered the functionalities attractive, the motivation for taking the courses was unclear. Additionally, the timeline needed to be more explicit about the steps, informative, and dynamic so that users could move through it easily.

After presenting the prototype and test results to the organization administrators, we agreed that a possible motivation could be a point-based reward system. This system would allow adopters to get points from courses and then redeem those points for services or products for their pets. The organization deemed it feasible to negotiate such discounts with their different supporting partners.

5.3 Second Iteration: Higher-Fidelity Prototype

Using the feedback from the paper prototype testing sessions, we moved to develop a higher fidelity prototype using Figma. This process entailed two design and testing cycles with six users per cycle. The cycle's main goal was to ensure that the app was perceived as useful and easy to navigate. At the end of both iteration and testing cycles, we put together a video prototype that we then showed to 23 potential adopters during an adoption fair. See fig2.b. After showing them the video, we asked them to fill out a form about their impressions of the app's usefulness.

The results indicated that the timeline was the functionality users perceived most useful. Thus, it needed to have a more prominent role in the app. Users also indicated they highly appreciated the product and service rewards the app could offer them. Finally, many mentioned they would like to continue using the timeline even after receiving the pet at home to know when vaccinations and other medical check-ups will need to take place.

After presenting the results to the organization, we brainstormed with them on making the timeline more visible and valuable. We decided to make the timeline the first screen of the app while also extending the timeline to include more vaccination and de-worming schedules as part of the milestones. Based on the participants' appreciation of possible rewards, we also decided to make the adopter's total reward points visible at all times.

Finally, the organization evaluated the feasibility of having volunteers address adopters' questions if a chatbot cannot offer support. They decided that, for now, it was not a feasible route due to their limited human infrastructure and chose to provide instead a Frequently Asked Questions (FAQ) section for adopters to browse.

6 AGAPET: THE LATEST ITERATION

The final version of the app included all the obtained feedback from previous cycles and usability heuristics. To make it accessible for users, an organization using Agapet will suggest users install it once they have filled out the application form. It will also provide users a code for registering in the app and linking to their adoption process. The app will meet the design implications via the following mechanisms:

6.1 Visibility of Status and Expectations

Once the adopter logs in, the first screen they will see is the adoption process timeline with all its phases and with a clear view of where they are in the process. They will receive notifications and information about the next stage as the process progresses. Once the adoption application is accepted, the timeline will show other milestones regarding medical checkups and vaccination schedules for the newly adopted pet. See fig3.a and b

The app also enables users to access their pet's medical record, including the status of vaccination updates and surgical interventions, to maximize support for adopters as they move forward with their pets. See fig4.c.

6.2 Learning About Responsible Adoption

While the adopter waits for the next phase in the adoption process, the app will recommend short classes about responsible adoption.

Courses are short and divided into brief video capsules. See fig5.a and b.

To further motivate users to take the courses, the app offers points that adopters can redeem for discounts for services and products in a section where they browse and look for various offers and discounts. Once the user chooses an offer to redeem their points, a member of the organization's staff receives an email and contacts the user to coordinate better details for accessing the service or product. The adopter's total points are always kept visible across screens. The app will also issue periodic notifications to users about new offers. See fig5.a and b and fig6.a.

6.3 Immediate Assistance

If having doubts about the adoption process and responsible adoption, users will be able to access a FAQ. See fig6.b



Figure 3: Agapet's functionalities: a) Timeline and b) Notifications

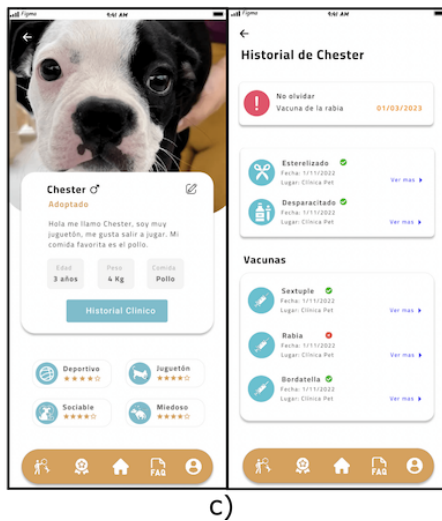


Figure 4: Agapet's functionalities: b) Pet's medical history



Figure 5: Agapet's functionalities: a) Available courses and b) Course Test

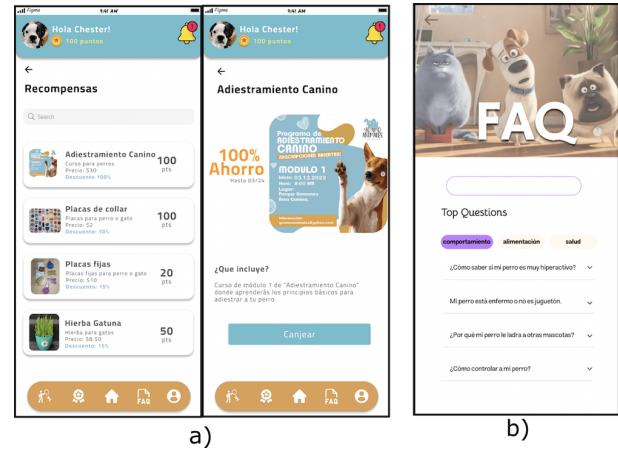


Figure 6: Agapet's functionalities: a) Available Rewards and b) FAQ

7 EVALUATION AND FUTURE WORK

We evaluated the last version of our high-fidelity prototype with 25 students from a Human-Computer Interaction course at our university. Our goal was to assess the previous three critical additions to the prototype (medical history and milestones, timeline prominence, and after-adoption extensions) in terms of their discoverability and perceived usefulness. To that end, we asked users to perform four tasks while we measured task completeness and demanded time: (a) to find the medical history of the pet; (b) to take a course that they browse and complete the test; (c) to redeem reward points, and (d) to check their status in the adoption process.

At the end of the test, we ask users to assess their satisfaction with the app's attractiveness, creativity, and efficiency using an AttrakDiff Questionnaire [11].

The completion rate was 100%, and all tasks took less than 30 seconds for users to complete: On average, Tasks 1 and 2 took 23 and 21 seconds, respectively; Tasks 3 and 4 took 12 seconds. Further,

77% of users answered efficiency as the main attribute of the app. That is, most users consider that the app allows them to attain their goals.

However, the high amount of time users invested in finding the pet's medical history and taking a course suggests those two areas need to be revised. In particular, the route to finding the medical record could have been clearer for many; once they clicked on the pet's picture, they expected to get to the pet's medical history immediately. In terms of taking a course, users took longer to figure out which menu icon led them to this functionality. Users suggested finding a more related icon. Finally, some users indicated the reward functionality needs a way for adopters to check on the history of their redeemed rewards.

We presented these results to our partner organization, and they have decided to move forward with implementing the app—with the recommended changes—through one of our university's computer science capstone classes. Their goal is to have it ready for deployment by late 2023. The organization sees excellent potential for the app to allow them to have a closer look at the pool of potential adopters and learn more about how to present educational resources to this population to maximize responsible adoption. The organization staff has expressed that the reward system can help them increase the available resources they already have. In securing discount agreements that will attract more clients, businesses will be more willing to learn about and support organizations in their efforts to save stray animals. Finally, YoAmoALosAnimales sees this app as one that can be helpful for more organizations in Guayaquil and other areas of the world. The organization hopes to explore business models that enable it to harness the competitive advantage it would have if it released the app to the public before others.

8 CONCLUSION

In finalizing this year-long project, we have learned that tight-knit work with organizations that support larger communities such as YoAmoALosAnimales, a willingness to test—both in in-the-wild and more controlled contexts. Further, iterating is critical for designing an application for already invested stakeholders who seek to work towards socially-relevant goals. We also learned that even though incorporating modern, state-of-the-art features such as a chatbot can better support users, their feasibility, when designing for real-world demands and limitations, cannot be taken for granted. Our evaluation of Agapet and its key features, such as a timeline, pet ownership short courses, and a reward system, suggest adopters will find it helpful in keeping up to date with their adoption process and decreasing uncertainty levels. If more potential adopters feel they can successfully engage in the adoption process, less strayed and abandoned animals will populate urban areas. Thus, responsible adoption can help cities attain the SDG 11 of fostering sustainable cities and communities. Overall, the organization we worked with and ourselves believe this app is a first step towards carefully introducing technology in pet adoption.

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